

STAT 412 HW07 — Question 11

Problem

In the American Heart Association journal *Hypertension*, researchers report that individuals who practice Transcendental Meditation (TM) lower their blood pressure significantly. If a random sample of 240 male TM practitioners meditate for 10.5 hours per week with a standard deviation of 2.22 hours, does that suggest that, on average, men who use TM meditate more than 10 hours per week? Quote a P-value in your conclusion.

Solution

The phrase “more than 10 hours” tells us the alternative is right-tailed:

$$H_0 : \mu = 10, \quad H_1 : \mu > 10.$$

So the H_0 dropdown is “equals,” and the H_1 dropdown is “greater than.” This is not a “not equal” test.

Because the sample standard deviation is used, use a t statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{10.5 - 10}{2.22/\sqrt{240}} \approx 3.4892.$$

With $df = 240 - 1 = 239$, the right-tail P-value is

$$P(T_{239} > 3.4892) \approx 0.0003.$$

If the platform asks for three decimals, enter 0.000.

In general, reject H_0 for significance levels greater than the P-value.

Final Answer

$H_0 : \mu = 10$; $H_1 : \mu > 10$. Test statistic $t = 3.49$. P-value ≈ 0.0003 ; reject H_0 . There is sufficient evidence that male TM practitioners meditate more than 10 hours per week on average. Reject for significance levels greater than the P-value.